

Theory of Computations

SOLUTIONS $\Rightarrow$

Name = Suman Kushwaha

Roll = 03

Semester = 4th

Subject = Theory of Computations

GitHub = Kushwaha-Suman

Website = Kushwahasuman.com.hp

Unit - wise Past Year Solutions:

Kleene Closure (L^*)

→ The Kleene closure of a language L is the set of all strings formed by concatenating zero or more strings from L .

$$L^* = \lambda \in \lambda \cup L^1 \cup L^2 \cup L^3 \cup \dots$$

where ϵ is the empty string

Note: L^* contains the empty string (ϵ), while L^+ does not.

(2) Differentiate Kleene closure from positive closure.
compute positive and Kleene closure of $\lambda ab \lambda$.
(2080 new)

→

Kleene Closure (L^*)

→ zero or more concatenations of strings from L

→ contains the empty string (ϵ)

$$L^* = \lambda \epsilon \lambda \cup L^+$$

problem

$$L^+ = L \cdot L^*$$

Positive Closure (L^+)

one or more concatenations of strings from L

→ does not contain empty string (ϵ)

$$L^+ = L \cdot L^*$$

positive closure (L^+)

$$L^+ = \lambda ab, abab, ababab, abababab, \dots$$

(one or more occurrences of "ab")

Kleen Closure (L^*)

$$L^* = \{ \epsilon, ab, abab, ababab, abababab, \dots \}$$

(zero or more occurrences of ab)

- ③ Define String, Substring, empty string and empty language over alphabet $\{a, b\}$

2080

Let the alphabet be

$$\Sigma = \{a, b\}$$

1 String:

→ A string is a finite sequence of symbol taken from an alphabet.

e.g. a, b, ab, aab, abb are strings over $\{a, b\}$

2 Substring:

→ A sub-string is a contiguous sequence of characters occurring within a string.

e.g. for the string $abba$, bb, ab and ba are substring of $abba$.

3) Empty String (ϵ):

→ The empty string is a string that contains no symbols. Its length is zero.

$$\text{e.g. } \epsilon \quad |\epsilon| = 0$$

4 Empty language ($\emptyset$):

→ The empty language is a language that contains no strings at all.

$$\emptyset = \{\}$$

④ Define the term alphabet, prefix and suffix of string, concatenation and Kleen closure with example, (2078)

Soln, Let alphabet are

$$\Sigma = \{a, b\}$$

1 Alphabet

→ An alphabet is a finite, non-empty set of symbols used to form strings.

Eg $\Sigma = \{a, b\}$

Strings like a, b, ab, ba are formed using this alphabet.

2 prefix of a string:

→ A prefix is any initial part of a string, including the empty string and the string itself.

Eg for string abba, prefix are

$$\epsilon, a, ab, abb, abba$$

3) Suffix of a string:

→ A suffix is any ending part of a string, including the empty string and the string itself.

e.g. for string abba, suffixes are:

$\epsilon, a, ba, bba, abba$

4) Concatenation:

→ concatenation is the operation of joining two strings end to end.

e.g. if $x = ab$, $y = ba$ then
 $xy = abba$

5) Kleene Closure (L^*)

→ Kleene closure of a language L is the set of all strings formed by concatenating zero or more strings from L .

$$L^* = \{\epsilon\} \cup L \cup L^2 \cup L^3 \cup \dots$$

e.g. if $L = \{ab\}$ then

$$L^* = \{\epsilon, ab, abab, ababab, \dots\}$$

Key Point :

- Prefix = Starting part of string
- Suffix = End part of string
- Concatenation = Joining string
- Kleene Closure = Repetition of language including empty string
- Alphabet = Set of symbols used to build strings.